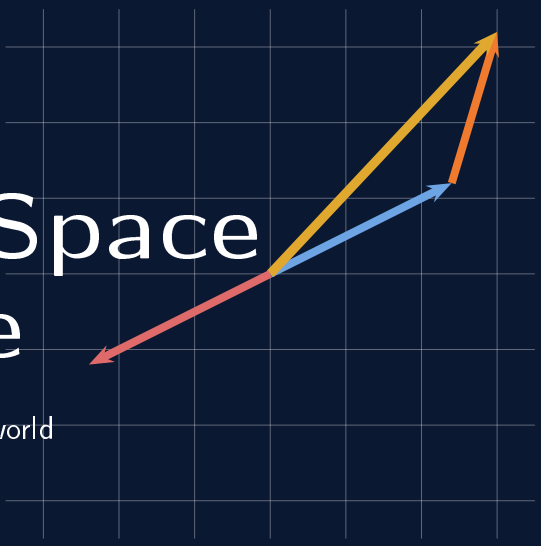
ZMATHLAB • LINEAR ALGEBRA

Vectors in 2-Space, 3-Space and n-Space



Size and direction: the language of the world

Lecture notes for BS students

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Based on: H. Anton, C. Rorres & A. Kaul, *Elementary Linear Algebra*, Section 3.1

What you will be able to do after these notes

- tell a scalar from a vector, and explain *why* direction matters;
- add, subtract and scale vectors **geometrically** (triangle and parallelogram rules);
- work with vectors in coordinates in $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$;
- find the vector between two points (*end minus start*);
- use the eight algebraic rules and build linear combinations;
- avoid the classic student mistakes, and solve every problem of Exercise Set 3.1.

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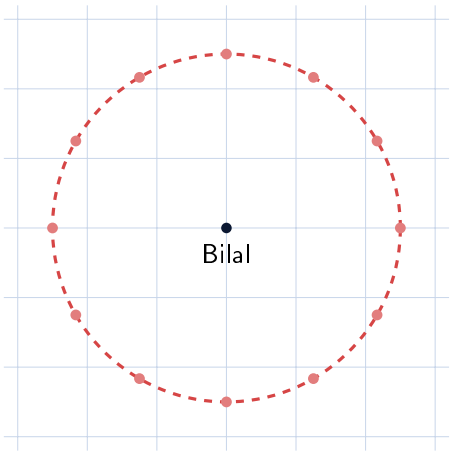
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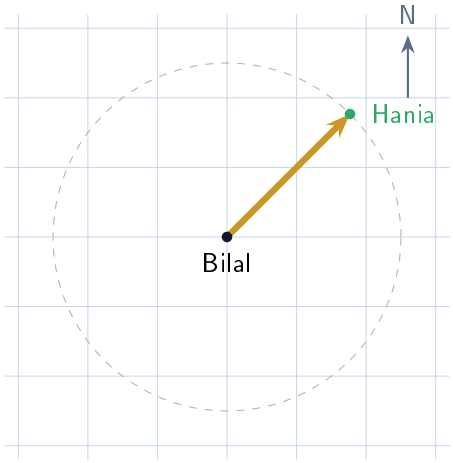
1 The story: lost at the mela

Story

Eid mela. Thousands of people, music, balloons. And Bilal's five-year-old daughter, Hania, is gone.
He grabs a stall keeper: *"Have you seen a little girl in a yellow dress?"* The man nods: *"Yes. She is **50 steps** away."*
Fifty steps. Bilal turns around, and around. Fifty steps **which way**? Hania could be at *any* point of a whole circle.
Then the man points: *"That way. Fifty steps, **north-east**."* Thirty seconds later, Hania is in her father's arms.



"50 steps"
a number only: she could be anywhere on the circle



"50 steps, north-east"
number + direction: exactly one place

In simple words

A **number** tells you *how much*. A **vector** tells you *how much* **and** *which way*. That one extra piece of information, the direction, is the whole reason vectors exist.

2 Scalars and vectors

Engineers and physicists divide physical quantities into two types.

	Scalar	**Vector**
Needs	a number only ("how much")	a number *and* a direction ("how much" and "which way")
Examples	temperature 20°C, length 5 cm, speed 75 km/h, mass, time, area	velocity (10 knots, 45° north-east), force (100 lb, vertically down), displacement, acceleration
Notation	lowercase italic: a, k, v, x	boldface: $\mathbf{a}$, $\mathbf{v}$, $\mathbf{w}$, $\mathbf{x}$ (handwritten: $\vec{v}$ or $\underline{v}$)

Teacher's tip

Speed vs velocity is the classic pair: a car doing 60 km/h has a *speed* (scalar). The same car doing 60 km/h *towards Islamabad* has a *velocity* (vector).

3 Where vectors live: applications

Navigation & GPS

Google Maps never says only "2 km". It says "2 km, *turn left*". Every step of a route is a displacement vector, and the whole trip is their sum.

Aviation: pilot and wind

The plane's velocity relative to the air plus the wind's velocity gives the true velocity over the ground. Forget the wind vector and you land in the wrong city.

Sport

A footballer's pass or a cricketer's shot: the ball needs the right speed *and* the right direction.

Computer graphics & games

Every movement of a game character is a vector added to its position, many times a second.

Engineering

The forces on a bridge (weight, cable tension, wind) are vectors; the bridge stands only if they add to zero.

Colour on screens

Each colour is a vector (r, g, b) of red, green, blue amounts (Section 14.1).

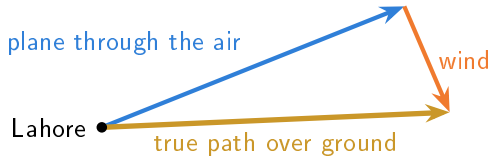
Data, economics, AI

A list of n measurements is a vector in $\mathbb{R}^n$: experiment data, trucks in 15 depots, outputs of 10 economic sectors, and the "word vectors" used in AI.

Physics

Einstein's space-time is 4-dimensional; string theory works in 11 dimensions. Vectors with many components describe them.

Pilot and wind: true velocity = air velocity + wind

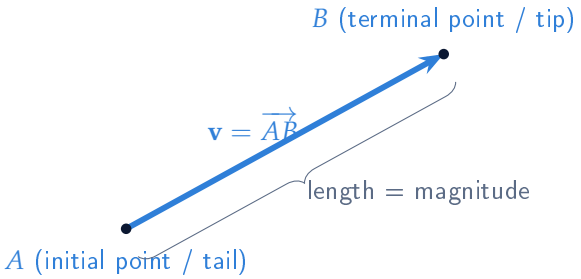


4 Geometric vectors

Definition: Geometric vector

In 2-space (the plane) or 3-space, a vector is represented by an **arrow**. The **length** of the arrow is the *magnitude*; the **arrowhead** gives the *direction*. The tail is the **initial point** and the tip is the **terminal point**. If the arrow goes from A to B we write

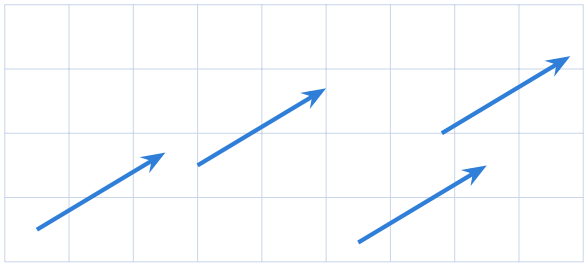
$$\mathbf{v} = \overrightarrow{AB}.$$



4.1 Equivalent (equal) vectors

Definition: Equivalent vectors

Vectors with the **same length** and the **same direction** are called *equivalent*. Because a vector is determined only by its length and direction, equivalent vectors are regarded as the **same vector**, even when they sit in different (parallel) positions. We write $\mathbf{v} = \mathbf{w}$.



four arrows, one vector: same length, same direction

In simple words

A vector is like an instruction ("2 right, 1.2 up"). Where you start following the instruction does not change the instruction.

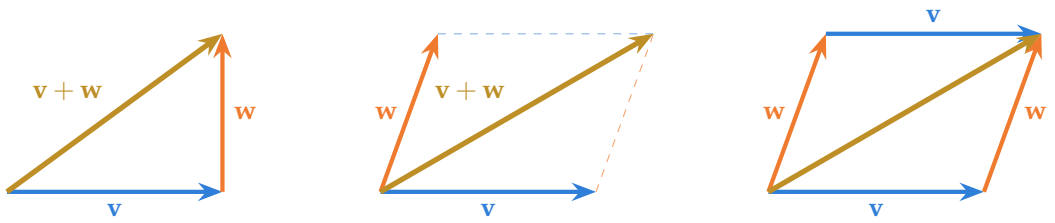
4.2 The zero vector

The vector whose initial and terminal points coincide has length zero. It is the **zero vector 0**. It has no natural direction, so we agree it may be given *any* direction that is convenient.

5 Vector addition

Example: Back to the mela

Hania walks 3 steps east ($\mathbf{v}$) and then 4 steps north ($\mathbf{w}$). Her total displacement is $\mathbf{v}+\mathbf{w}$: an arrow from where she started to where she stopped, 5 steps long (by Pythagoras, $\sqrt{3^2+4^2}=5$).



(a) Triangle rule (b) Parallelogram rule (c) $\mathbf{v}+\mathbf{w}=\mathbf{w}+\mathbf{v}$

Parallelogram rule

If $\mathbf{v}$ and $\mathbf{w}$ are positioned so that their initial points coincide, they form adjacent sides of a parallelogram, and $\mathbf{v}+\mathbf{w}$ is the arrow from the common initial point to the *opposite* vertex.

Triangle rule

If the initial point of $\mathbf{w}$ is placed at the terminal point of $\mathbf{v}$, then $\mathbf{v}+\mathbf{w}$ is the arrow from the initial point of $\mathbf{v}$ to the terminal point of $\mathbf{w}$.

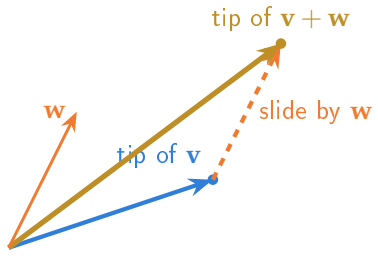
Figure (c) shows that both orders reach the same corner, so

$$\boxed{\mathbf{v}+\mathbf{w}=\mathbf{w}+\mathbf{v}} \qquad \text{(commutative law)}$$

and that the triangle rule and the parallelogram rule give the *same* vector.

5.1 Addition viewed as translation

If $\mathbf{v}$, $\mathbf{w}$ and $\mathbf{v}+\mathbf{w}$ all start at the same point, the tip of $\mathbf{v}+\mathbf{w}$ is obtained by **sliding (translating)** the tip of $\mathbf{v}$ in the direction of $\mathbf{w}$ by the length of $\mathbf{w}$ (or the tip of $\mathbf{w}$ along $\mathbf{v}$). So $\mathbf{v}+\mathbf{w}$ is "the translation of $\mathbf{v}$ by $\mathbf{w}$".

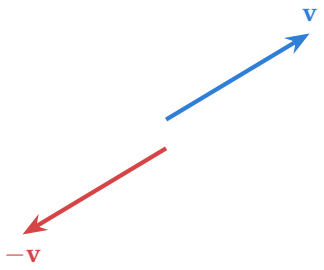


6 Vector subtraction

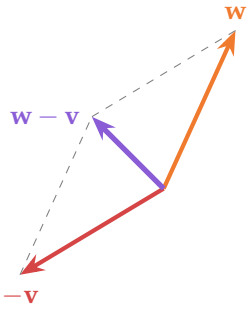
Definition: Negative and difference

The **negative** of $\mathbf{v}$, written $-\mathbf{v}$, has the *same length* as $\mathbf{v}$ but the *opposite direction*. The **difference** of $\mathbf{v}$ from $\mathbf{w}$ is

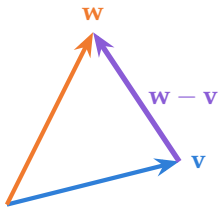
$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v}).$$



(a) $\mathbf{v}$ and $-\mathbf{v}$



(b) $\mathbf{w} + (-\mathbf{v})$



(c) quick way: from tip of $\mathbf{v}$ to tip of $\mathbf{w}$

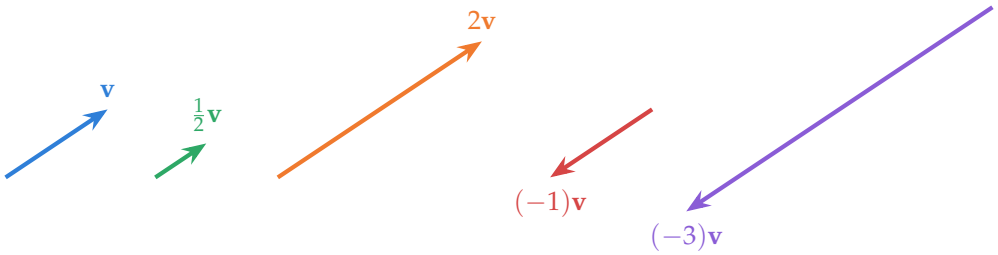
Teacher's tip

Memory trick: $\mathbf{w} - \mathbf{v}$ is the arrow that you must *add to* $\mathbf{v}$ to *get* $\mathbf{w}$ (because $\mathbf{v} + (\mathbf{w} - \mathbf{v}) = \mathbf{w}$). So it starts at $\mathbf{v}$'s tip and ends at $\mathbf{w}$'s tip.

7 Scalar multiplication

Definition: Scalar multiple

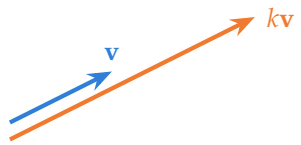
If $\mathbf{v} \neq \mathbf{0}$ and $k \neq 0$, then $k\mathbf{v}$ is the vector whose length is $|k|$ times the length of $\mathbf{v}$, and whose direction is the *same* as $\mathbf{v}$ if $k > 0$ and *opposite* if $k < 0$. If $k = 0$ or $\mathbf{v} = \mathbf{0}$, then $k\mathbf{v} = \mathbf{0}$.



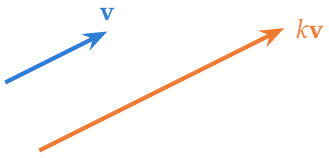
In particular $(-1)\mathbf{v} = -\mathbf{v}$.

7.1 Parallel and collinear vectors

If one of two vectors is a scalar multiple of the other ($\mathbf{w} = k\mathbf{v}$), they lie along a common line: they are **collinear**. If we translate one of them they look "parallel but not collinear". Since translating does not change a vector, for vectors the words **parallel** and **collinear** mean the same thing. The zero vector is regarded as parallel to every vector.



(a) collinear



(b) translated: still parallel

Example: Is $(6,-9)$ parallel to $(-2,3)$?

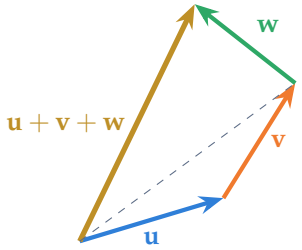
We look for k with $(6,-9) = k(-2,3)$. First components: $6 = -2k \Rightarrow k = -3$. Check the second: $3k = 3(-3) = -9$ ✓. So yes, $(6,-9) = -3(-2,3)$: parallel, opposite direction, three times as long.

8 Sums of three or more vectors

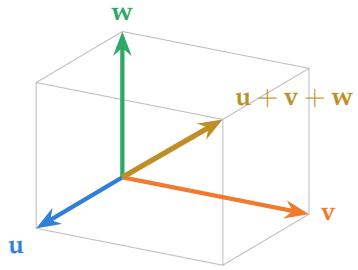
Vector addition is **associative**:

$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w},$$

so $\mathbf{u} + \mathbf{v} + \mathbf{w}$ has only one meaning. Place the vectors **tip to tail** and draw the arrow from the first tail to the last tip. In 3-space, if $\mathbf{u}, \mathbf{v}, \mathbf{w}$ start at one point, then $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the **diagonal of the parallelepiped (box)** with those three edges.



tip to tail



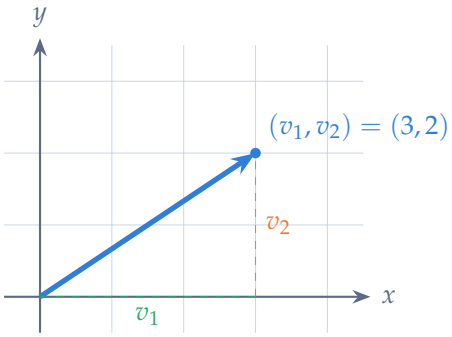
diagonal of the box

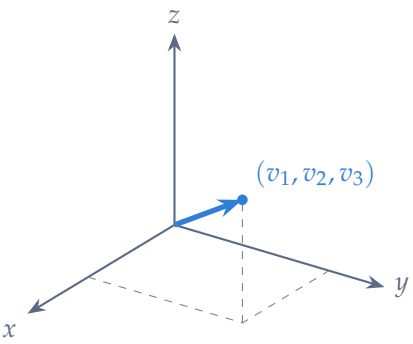
9 Vectors in coordinate systems

Computations are much easier on a coordinate grid. Put the **initial point at the origin**; then the vector is completely determined by the coordinates of its terminal point. These are the **components** of the vector:

$$\mathbf{v} = (v_1, v_2) \text{ in 2-space}, \qquad \mathbf{v} = (v_1, v_2, v_3) \text{ in 3-space}.$$

The zero vector is $\mathbf{0} = (0,0)$ in 2-space and $\mathbf{0} = (0,0,0)$ in 3-space.





Equality in components

Two vectors are equivalent *if and only if* their corresponding components are equal: $(v_1, v_2, v_3) = (w_1, w_2, w_3) \iff v_1 = w_1,\ v_2 = w_2,\ v_3 = w_3$.

Teacher's tip

Point or vector? The pair (v_1, v_2) can mean the *point* with those coordinates, or the *arrow* from the origin to that point. Both readings are correct; the problem tells you which picture to use.

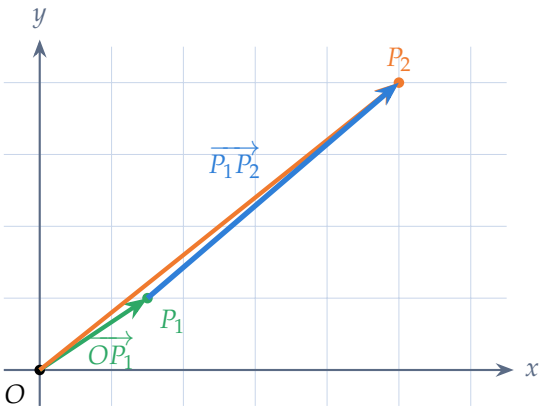
10 Vectors whose initial point is not at the origin

If $\overrightarrow{P_1P_2}$ goes from $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$, then (Figure below)

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = (x_2 - x_1,\ y_2 - y_1),$$

and in 3-space, from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$:

$$\boxed{\overrightarrow{P_1P_2} = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)} \qquad \textbf{end minus start.}$$



Example: Anton, Example 1 (every step)

$P_1(2, -1, 4)$, $P_2(7, 5, -8)$. End minus start, component by component:

$$\begin{aligned} x:&\quad 7 - 2 = 5 \\ y:&\quad 5 - (-1) = 5 + 1 = 6 \\ z:&\quad (-8) - 4 = -12 \end{aligned}$$

So $\mathbf{v} = \overrightarrow{P_1P_2} = (5,\ 6,\ -12)$.

11 n-space

The real line is $\mathbb{R} = \mathbb{R}^1$ (one-dimensional), the plane is $\mathbb{R}^2$ (ordered pairs), and space is $\mathbb{R}^3$ (ordered triples). Why stop at 3?

Definition: Ordered n-tuple and $\mathbb{R}^n$

If n is a positive integer, an **ordered n-tuple** is a sequence of n real numbers $(v_1, v_2, \ldots, v_n)$. The set of all ordered n-tuples is called **real n-space**, written $\mathbb{R}^n$.

You may picture $(v_1, \ldots, v_n)$ as a generalized *point* or as a generalized *vector*; mathematically it makes no difference, because what matters is the algebra.

Typical n-tuples (from Anton)

- **Experimental data:** n measurements per experiment give $\mathbf{y} = (y_1, \ldots, y_n) \in \mathbb{R}^n$.
- **Storage and warehousing:** trucks in 15 depots give $\mathbf{x} = (x_1, \ldots, x_{15}) \in \mathbb{R}^{15}$.
- **Electrical circuits:** a chip with 4 input and 3 output voltages turns a vector in $\mathbb{R}^4$ into a vector in $\mathbb{R}^3$.
- **Graphical images:** a pixel is (x, y, h, s, b): position, hue, saturation, brightness, a vector in $\mathbb{R}^5$.
- **Economics:** the outputs of 10 sectors form $\mathbf{s} = (s_1, \ldots, s_{10}) \in \mathbb{R}^{10}$.
- **Mechanical systems:** 6 positions, 6 velocities and time form the *state* vector in $\mathbb{R}^{13}$.

12 Operations on vectors in $\mathbb{R}^n$

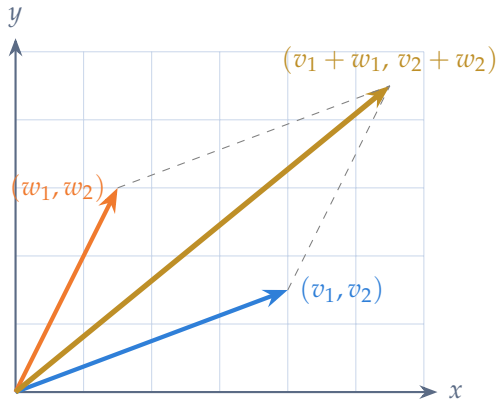
Definition: Equality

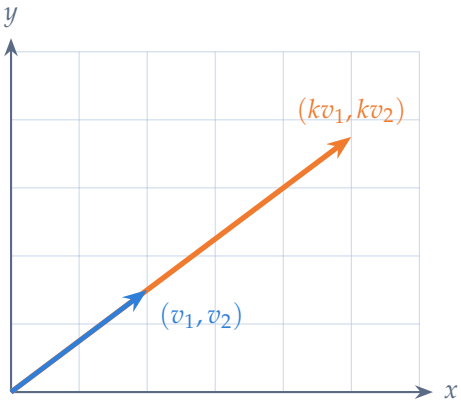
$\mathbf{v} = (v_1, \ldots, v_n)$ and $\mathbf{w} = (w_1, \ldots, w_n)$ are **equal** if $v_1 = w_1,\ v_2 = w_2,\ \ldots,\ v_n = w_n$.

Example: Anton, Example 2

$(a, b, c, d) = (1, -4, 2, 7)$ if and only if $a = 1,\ b = -4,\ c = 2,\ d = 7$.

The picture in $\mathbb{R}^2$ shows why the operations are defined componentwise:





Definition: Operations in $\mathbb{R}^n$

For $\mathbf{v} = (v_1, \ldots, v_n)$, $\mathbf{w} = (w_1, \ldots, w_n)$ and a scalar k:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1,\ v_2 + w_2,\ \ldots,\ v_n + w_n) \tag{10}$$
$$k\mathbf{v} = (kv_1,\ kv_2,\ \ldots,\ kv_n) \tag{11}$$
$$-\mathbf{v} = (-v_1,\ -v_2,\ \ldots,\ -v_n) \tag{12}$$
$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v}) = (w_1 - v_1,\ w_2 - v_2,\ \ldots,\ w_n - v_n) \tag{13}$$

In simple words

Add (or subtract) vectors by adding (or subtracting) *matching* components. Multiply by a scalar by multiplying *every* component.

Example: Anton, Example 3 (every step)

$\mathbf{v} = (1, -3, 2)$, $\mathbf{w} = (4, 2, 1)$.

$$\mathbf{v} + \mathbf{w} = (1+4,\ -3+2,\ 2+1) = (5,\ -1,\ 3)$$
$$2\mathbf{v} = (2\cdot 1,\ 2\cdot(-3),\ 2\cdot 2) = (2,\ -6,\ 4)$$
$$-\mathbf{w} = (-4,\ -2,\ -1)$$
$$\mathbf{v} - \mathbf{w} = \mathbf{v} + (-\mathbf{w}) = (1-4,\ -3-2,\ 2-1) = (-3,\ -5,\ 1)$$

13 The algebraic rules

Theorem 3.1.1

If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and k, m are scalars, then

(a)	$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$	order does not matter
(b)	$(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$	grouping does not matter
(c)	$\mathbf{u}+\mathbf{0}=\mathbf{0}+\mathbf{u}=\mathbf{u}$	adding zero changes nothing
(d)	$\mathbf{u}+(-\mathbf{u})=\mathbf{0}$	every vector has an opposite
(e)	$k(\mathbf{u}+\mathbf{v})=k\mathbf{u}+k\mathbf{v}$	a scalar distributes over a sum
(f)	$(k+m)\mathbf{u}=k\mathbf{u}+m\mathbf{u}$	a sum of scalars distributes
(g)	$k(m\mathbf{u})=(km)\mathbf{u}$	scaling twice = scaling once by the product
(h)	$1\mathbf{u}=\mathbf{u}$	multiplying by 1 changes nothing

Proof of (b). Let $\mathbf{u}=(u_1,\dots,u_n)$, $\mathbf{v}=(v_1,\dots,v_n)$, $\mathbf{w}=(w_1,\dots,w_n)$. Then

$$\begin{aligned}
(\mathbf{u}+\mathbf{v})+\mathbf{w} &= (u_1+v_1,\dots,u_n+v_n)+(w_1,\dots,w_n) && \text{[vector addition]}\\
&= ((u_1+v_1)+w_1,\dots,(u_n+v_n)+w_n) && \text{[vector addition]}\\
&= (u_1+(v_1+w_1),\dots,u_n+(v_n+w_n)) && \text{[regroup real numbers]}\\
&= (u_1,\dots,u_n)+(v_1+w_1,\dots,v_n+w_n) && \text{[vector addition]}\\
&= \mathbf{u}+(\mathbf{v}+\mathbf{w}). \qquad \blacksquare
\end{aligned}$$

Theorem 3.1.2

For $\mathbf{v}\in\mathbb{R}^n$ and a scalar k: (a) $0\mathbf{v}=\mathbf{0}$, (b) $k\mathbf{0}=\mathbf{0}$, (c) $(-1)\mathbf{v}=-\mathbf{v}$.

13.1 Calculating without components

The rules let us solve vector equations without ever writing components. Solve $\mathbf{x}+\mathbf{a}=\mathbf{b}$:

$$\begin{aligned}
\mathbf{x}+\mathbf{a} &= \mathbf{b} && \text{[given]}\\
(\mathbf{x}+\mathbf{a})+(-\mathbf{a}) &= \mathbf{b}+(-\mathbf{a}) && \text{[add } -\mathbf{a} \text{ to both sides]}\\
\mathbf{x}+(\mathbf{a}+(-\mathbf{a})) &= \mathbf{b}-\mathbf{a} && \text{[Theorem 3.1.1(b)]}\\
\mathbf{x}+\mathbf{0} &= \mathbf{b}-\mathbf{a} && \text{[Theorem 3.1.1(d)]}\\
\mathbf{x} &= \mathbf{b}-\mathbf{a} && \text{[Theorem 3.1.1(c)]}
\end{aligned}$$

Teacher's tip

These eight rules are exactly the rules that define a **vector space**. Anything that obeys them (polynomials, matrices, signals, images) can be treated with the same tools.

14 Linear combinations

Definition: Linear combination

$\mathbf{w}\in\mathbb{R}^n$ is a **linear combination** of $\mathbf{v}_1,\dots,\mathbf{v}_r$ if

$$\mathbf{w}=k_1\mathbf{v}_1+k_2\mathbf{v}_2+\cdots+k_r\mathbf{v}_r \tag{14}$$

for some scalars $k_1,\dots,k_r$, called the **coefficients**. With $r=1$, a linear combination is just a scalar multiple $k_1\mathbf{v}_1$.

Example: Building $(7,4)$ from $(1,2)$ and $(3,0)$

We want $k_1(1,2)+k_2(3,0)=(7,4)$, i.e. $(k_1+3k_2,\ 2k_1)=(7,4)$. Second component: $2k_1=4\Rightarrow k_1=2$. First component: $2+3k_2=7\Rightarrow 3k_2=5\Rightarrow k_2=\frac{5}{3}$. Check: $2(1,2)+\frac{5}{3}(3,0)=(2,4)+(5,0)=(7,4)$ ✓.

14.1 Application: the RGB colour cube

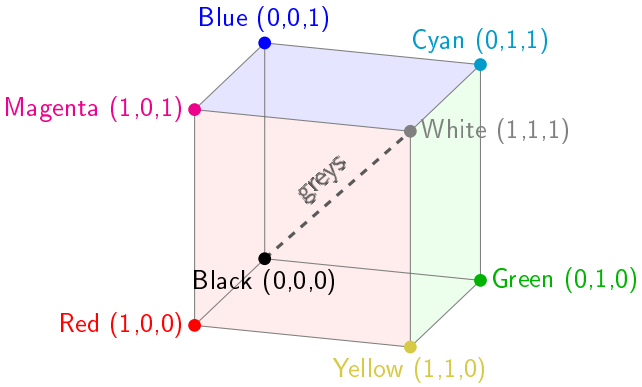
Screens mix red, green and blue light. Identify the pure colours with

$$\mathbf{r}=(1,0,0),\qquad \mathbf{g}=(0,1,0),\qquad \mathbf{b}=(0,0,1).$$

Every colour is a linear combination

$$\mathbf{c}=k_1\mathbf{r}+k_2\mathbf{g}+k_3\mathbf{b}=(k_1,k_2,k_3),\qquad 0\le k_i\le 1,$$

so all colours fill the unit cube. Its corners are black, the three primaries, cyan, magenta, yellow and white; the diagonal from black to white holds the greys.



15 Alternative notations

A vector in $\mathbb{R}^n$ is an ordered list, so any notation that keeps the order is valid:

$$\underbrace{\mathbf{v}=(v_1,v_2,\ldots,v_n)}_{\text{comma-delimited}}\qquad \underbrace{\mathbf{v}=\begin{bmatrix}v_1 & v_2 & \cdots & v_n\end{bmatrix}}_{\text{row vector}}\qquad \underbrace{\mathbf{v}=\begin{bmatrix}v_1\\ v_2\\ \vdots\\ v_n\end{bmatrix}}_{\text{column vector}}$$

16 Common student misconceptions

Misconception 1: "If I move the arrow, I get a different vector"

Wrong: treating two equal-length, same-direction arrows at different places as different vectors.
Right: a vector is only length + direction. $\overrightarrow{AB}$ with $A(0,0), B(2,1)$ and $\overrightarrow{CD}$ with $C(5,3), D(7,4)$ are both $(2,1)$: the *same* vector.

Misconception 2: "start minus end"

Wrong: for $P_1(3,5)$, $P_2(2,8)$ writing $\overrightarrow{P_1P_2} = (3-2,\ 5-8) = (1,-3)$.
Right: **end minus start**: $(2-3,\ 8-5) = (-1,3)$. The wrong answer is $-\overrightarrow{P_1P_2} = \overrightarrow{P_2P_1}$.

Misconception 3: the direction of $\mathbf{w}-\mathbf{v}$

Wrong: drawing $\mathbf{w}-\mathbf{v}$ from the tip of $\mathbf{w}$ to the tip of $\mathbf{v}$.
Right: from the tip of $\mathbf{v}$ to the tip of $\mathbf{w}$. Check: $\mathbf{v}+(\mathbf{w}-\mathbf{v}) = \mathbf{w}$, so you travel along $\mathbf{v}$ and then along $\mathbf{w}-\mathbf{v}$ to reach $\mathbf{w}$'s tip.

Misconception 4: "$-2\mathbf{v}$ is shorter than $\mathbf{v}$"

Wrong: thinking the minus sign makes a vector smaller.
Right: length of $k\mathbf{v}$ is $|k|$ times the length of $\mathbf{v}$. $-2\mathbf{v}$ is *twice* as long; the minus sign only reverses the direction.

Misconception 5: adding lengths

Wrong: "3 steps east plus 4 steps north = 7 steps from the start".
Right: lengths add only when the vectors point the same way. Here $(3,0)+(0,4) = (3,4)$, whose length is $\sqrt{9+16} = 5$.

Misconception 6: point versus vector

Wrong: mixing up "the point $(2,3)$" and "the vector $(2,3)$" in a question about initial/terminal points.
Right: a point is a location; a vector is a displacement. The terminal point of $\mathbf{u} = (1,2)$ starting at $A(1,1)$ is $A+\mathbf{u} = (2,3)$, not $(1,2)$.

Misconception 7: parallel test with one component

Wrong: "$(8t,-2)$ is parallel to $(4,-1)$ because $8t$ can be a multiple of 4 for any t".
Right: the *same* k must work for *every* component: $-2 = k(-1) \Rightarrow k = 2$, then $8t = 2\cdot 4 \Rightarrow t = 1$ only.

17 More worked examples

Example: Terminal point

$\mathbf{u} = (3,-2,5)$ starts at $A(1,4,-1)$. Terminal point $B = A+\mathbf{u} = (1+3,\ 4+(-2),\ -1+5) = (4,\ 2,\ 4)$.

Example: Initial point

$\mathbf{u} = (2,-1)$ ends at $B(5,3)$. Since $B-A = \mathbf{u}$, $A = B-\mathbf{u} = (5-2,\ 3-(-1)) = (3,\ 4)$.

Example: A linear combination with numbers

$\mathbf{u} = (2,0,-1)$, $\mathbf{v} = (1,3,4)$. Then $3\mathbf{u} - 2\mathbf{v} = (6,0,-3) - (2,6,8) = (6-2,\ 0-6,\ -3-8) = (4,\ -6,\ -11)$.

Example: Midpoint and a point $\frac{1}{3}$ of the way

$P(1,2,3)$, $Q(7,8,-3)$. $\overrightarrow{PQ} = (6,6,-6)$. Midpoint $= P + \frac{1}{2}\overrightarrow{PQ} = (1+3,\ 2+3,\ 3-3) = (4,5,0)$. Point $\frac{1}{3}$ of the way $= P + \frac{1}{3}\overrightarrow{PQ} = (1+2,\ 2+2,\ 3-2) = (3,4,1)$.

18 One-page summary

Vectors in one page

- **Vector** = magnitude + direction (arrow: tail = initial point, tip = terminal point). **Scalar** = number only.
- **Equivalent** vectors: same length and direction $\Rightarrow$ same vector, wherever drawn. $\mathbf{0}$: length 0, any direction.
- **Add:** tip to tail (triangle) or diagonal of parallelogram. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$. Addition = translation.
- **Subtract:** $\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v})$ = arrow from tip of $\mathbf{v}$ to tip of $\mathbf{w}$.
- **Scale:** $k\mathbf{v}$ is $|k|$ times as long; same way if $k > 0$, opposite if $k < 0$; $0\mathbf{v} = \mathbf{0}$.
- **Parallel = collinear:** one vector is a scalar multiple of the other.
- **Components:** $\mathbf{v} = (v_1, \ldots, v_n)$; equal $\iff$ all components equal.
- **End minus start:** $\overrightarrow{P_1P_2} = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)$.
- **Componentwise:** $\mathbf{v} + \mathbf{w} = (v_i + w_i)$, $k\mathbf{v} = (kv_i)$, $\mathbf{w} - \mathbf{v} = (w_i - v_i)$.
- **Eight rules** (Theorem 3.1.1) + $0\mathbf{v} = \mathbf{0}$, $k\mathbf{0} = \mathbf{0}$, $(-1)\mathbf{v} = -\mathbf{v}$.
- **Linear combination:** $k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r$. RGB colours: $k_1\mathbf{r} + k_2\mathbf{g} + k_3\mathbf{b}$.

19 Practice

From Anton, Exercise Set 3.1: **1–12** (components and operations), **13–14** (vector equations), **15–16** (parallel vectors), **17–22** (linear combinations and systems), **23–24, 27** (midpoints and points on a segment), **28–29** (thinking questions). Full step-by-step solutions are in the companion file *Vectors_2D_3D_Solved_Exercises.pdf*.